Math 534 H – Spring 2026 – Lecture 7b.

Section 1.6 Classification of linear second order equations

We will go through the details for $(x,y) \in \mathbb{R}\times\mathbb{R} = \mathbb{R}^2$ but similar arguments extend to any dimension ($\mathbb{R}^d$).

Consider a second order linear PDE with constant coeff.

$$0 = a_{11} u_{xx} + a_{12} u_{xy} + a_{21} u_{yx} + a_{22} u_{yy} + \underbrace{a_1 u_x + a_2 u_y + a_0 u}_{\text{l.o.t.}}$$

(lower order terms)

Assume $u \in C^2$ so that $\boxed{u_{xy} = u_{yx}}$

So we have $\underbrace{(a_{12} + a_{21})}_{:= a'_{12}} u_{xy}$

Abuse notation and label this number again by $2a_{12}$ $(= a'_{12})$

Next we ignore the l.o.t as will play no role.

So all in all we have

$$0 = a_{11} u_{xx} + 2a_{12} u_{xy} + a_{22} u_{yy}$$

Assume $a_{11} \neq 0$.

$$0 = u_{xx} + 2\frac{a_{12}}{a_{11}} u_{xy} + \frac{a_{22}}{a_{11}} u_{yy}$$

$$0 = \underbrace{\left(\partial_{xx} + 2\frac{a_{12}}{a_{11}}\partial_{xy} + \frac{a_{22}}{a_{11}}\partial_{yy}\right)}_{=L} u$$

$$= \partial_{xx} + 2\frac{a_{12}}{a_{11}}\partial_{xy} + \frac{a_{12}^2}{a_{11}^2}\cdot\partial_{yy} - \frac{a_{12}^2}{a_{11}^2}\partial_{yy} + \frac{a_{22}}{a_{11}}\partial_{yy}$$

$$= \underbrace{\left(\partial_x + \frac{a_{12}}{a_{11}}\partial_y\right)^2} + \underbrace{\left(\frac{a_{22}}{a_{11}} - \frac{a_{12}^2}{a_{11}^2}\right)}_{\Delta}\partial_{yy}$$

The classification depends on the sign of Δ

<u>Three cases</u> : $\Delta > 0$; $\Delta < 0$; $\Delta = 0$

1. $\Delta > 0 \iff \dfrac{a_{22}a_{11} - a_{12}^2}{a_{11}^2} > 0 \iff$

$a_{11}^2 > 0$ always!

$$\boxed{a_{22}a_{11} - a_{12}^2 > 0}$$

$\rightarrow$ EQUATION IS SAID TO BE <u>ELLIPTIC</u>

2. $\Delta < 0 \iff \boxed{a_{22}a_{11} - a_{12}^2 < 0}$ EQUATION IS <u>HYPERBOLIC</u>

3 If $\Delta = 0 \iff \boxed{a_{22}a_{11} - a_{12}^2 = 0}$

EQUATION IS SAID TO BE <u>PARABOLIC</u>

NEXT TIME WILL SHOW YOU WHY USING A CHANGE OF VARIABLES.

<u>Examples</u>: 1) $2u_{xx} + 5u_{yy} - 2u_{xy} + u_x = 0$

$\Delta = a_{22}a_{11} - a_{12}^2$ $\quad a_{11} = 2$ $\quad a_{12} = -1$

$a_{22} = 5$

So: $5 \cdot 2 - (-1)^2 = 10 - 1 = 9 > 0$

So Eq. is <u>ELLIPTIC</u>.

2) $u_{xx} - 6u_{xy} + 3u = 0$ $\quad a_{22} = 0$

$\Delta = 0 - (-3)^2 = -9 < 0$ $\quad a_{11} = 1$ $\quad a_{12} = -3$

HYPERBOLIC

3) $4u_{yy} + 4u_{xy} + u_{xx} + u_y - 1 = 0$

$$\underbrace{4u_{yy} + 4u_{xy} + u_{xx}}_{\text{Linear second order part.}} + u_y = \underbrace{1}_{\text{forcing term. (inhomog.)}}$$

$a_{11} = 1 \quad a_{22} = 4 \quad a_{12} = 2$

$\Delta = 4.1 - 2^2 = 0$ PARABOLIC.

THINK ABOUT VARIABLE COEFFICIENT:

4) $\underbrace{5y}\, u_{xx} - 8u_{xy} + \underbrace{2x}\, u_{yy} = 0$

$\Delta = (5y)(2x) - (-4)^2 = \underbrace{10xy - 8}_{\text{(NEXT TIME!)}}$.

When is $\Delta > 0$, $\Delta < 0$, $\Delta = 0$?